MATH 547: HOMEWORK 5
DUE ON: MONDAY, DECEMBER 2, 9AM.

Problem 1: Hoeffding's inequality in Hilbert spaces, 15 points:
Let $X_1, \dots, X_n$ be independent random vectors in a $\mathbb{R}^p$ such that $\mathbb{E}X_i = \vec{0},\ i = 1, \dots, n$ and $\|X_i\|_2 \le c/2$ (almost surely) for some $c > 0$, and set $B^2 := \frac{nc^2}{4}$. Then for all $t \ge B$,

$$\Pr\left(\left\|\sum_{i=1}^{n} X_i\right\|_2 \ge t\right) \le \exp\left(-\frac{(t-B)^2}{8B^2}\right).$$

[Hint: apply the bounded difference inequality to $\left\|\sum_{i=1}^n X_i\right\|_2$. Also, note that the result is valid for any separable Hilbert space $(\mathbb{H}, \langle\cdot,\cdot\rangle)$, not only $\mathbb{R}^p$.]

$\mathbb{E}X_i = 0 \quad \forall\, i \in \{1, \dots n\}$

$\|X_i\|_2 \le c/2$ for some $c > 0$

$B^2 := \frac{nc^2}{4}$

Bounded difference / McDiarmid's Inequality
(from Week 10 in lecture notes)

If $f(x_1, \dots x_n)$ is a function of n - variables & satisfies

$$\sup_{x_1, \dots x_j, x_j', \dots x_n} |f(x_1, \dots x_j, \dots x_n) - f(x_1, \dots x_j', \dots x_n)| \le c_j$$

for all $j \in \{1, \dots n\}$ & for some c_j, then

$$\Pr(|f(X_1 \dots X_n) - \mathbb{E}f(X_1, \dots, X_n)| \ge t) \le 2e^{-\frac{t^2}{2\sum_{j=1}^n c_j^2}}$$

We will use this inequality to prove the claim in the question.

Let $f(\vec{x}_1, \dots \vec{x}_n) = \left|\sum_{i=1}^n \vec{x}_i\right|_2$

$$|f(\vec{x}_1, \dots \vec{x}_j, \dots \vec{x}_n) - f(\vec{x}_1, \dots \vec{x}_j', \dots \vec{x}_n)|$$

$$= \Big| \underbrace{\Big|\sum_{i=1}^{n} \vec{x}_i\Big|}_{a} - \underbrace{\Big|\sum_{\substack{i=1 \\ i\neq j}}^{n} \vec{x}_i + \vec{x}_j'\Big|}_{b} \Big|$$

$$\leq \Big| \underbrace{\sum_{i=1}^{n} \vec{x}_i}_{a} - \underbrace{\sum_{\substack{i=1 \\ i\neq j}}^{n} \vec{x}_i + \vec{x}_j'}_{b} \Big| \quad \text{(reverse triangle inequality)}$$

$$= |\vec{x}_j - \vec{x}_j'| \leq |\vec{x}_j| + |\vec{x}_j'| \quad \text{(triangle ineq.)}$$

$$\leq c/2 + c/2 = c$$

$$\Rightarrow \sup_{x_1 \dots x_j, x_j', \dots x_n} |f(\vec{x}_1, \dots \vec{x}_j, \dots \vec{x}_n) - f(\vec{x}_1, \dots \vec{x}_j', \dots \vec{x}_n)| \leq c$$

* Hence the premise for McDiarmid's inequality is satisfied with $c_j = c \;\; \forall j \in \{1, \dots, n\}$

$$\sum_{i=1}^{n} c_i^2 = nc^2 = 4B^2$$

By McDiarmid's inequality,

$$\Pr\left(\Big|\Big|\sum_{i=1}^{n} x_i\Big| - \mathbb{E}\Big|\sum_{i=1}^{n} x_i\Big|\Big| \geq t\right) \leq 2\exp\left(\frac{-t^2}{2\sum_{i=1}^{n} c_i^2}\right)$$

$$= 2\exp\left(\frac{-t^2}{2nc^2}\right)$$

$$= 2\exp\left(\frac{-t^2}{8B^2}\right)$$

Let $S = \sum_{i=1}^{n} x_i$ → eq ①

Then, $\boxed{Pr(\,|\,|S| - \mathbb{E}|S|\,| \geq t) \leq 2\exp\left(\frac{-t^2}{8B^2}\right)}$

Also $(\mathbb{E}|S|)^2 \leq \mathbb{E}|S|^2$ (Jensen's inequality $(\cdot)^2$ is convex)

$$= \mathbb{E}\langle S, S\rangle$$

$$= \mathbb{E}\left\langle \sum_{i=1}^{n} x_i, \sum_{j=1}^{n} x_j \right\rangle$$

$$= \mathbb{E} \sum_{i=1}^{n}\sum_{j=1}^{n} \langle x_i, x_j\rangle$$

$$= \mathbb{E}\left(\sum_{i=1}^{n} |x_i|^2 + \sum_{i\neq j} \langle x_i, x_j\rangle \right)$$

$$= \mathbb{E}\sum_{i=1}^{n} |x_i|^2 + \sum_{i\neq j} \mathbb{E}\langle x_i, x_j\rangle$$

$\because x_i \& x_j$ are independent

$$= \sum_{i=1}^{n} \mathbb{E}|x_i|^2 + \sum_{i\neq j} \langle \underbrace{\mathbb{E}x_i}_{0}, \underbrace{\mathbb{E}x_j}_{0} \rangle$$

Since $|x_i|^2 \leq c^2/4$

$$\mathbb{E}|x_i|^2 \leq c^2/4$$

$$\sum_{i=1}^{n} \mathbb{E}|x_i|^2 \leq \frac{nc^2}{4} = B^2$$

$$\mathbb{E}|S|^2 \leq B^2$$

$$(E|s|)^2 \le E|s|^2 \le B^2$$

$$\Rightarrow E|s| \le B$$

For $t \ge B$, the event

$$\{|s| \ge t\} = \{|s| - E|s| \ge t - E|s|\}$$

$$\subseteq \{|s| - E|s| \ge t - B\}$$

$$\therefore \Pr\{|s| \ge t\} \le \Pr(|s| - E|s| \ge t - B)$$

$$\le 2\exp\left(-\frac{(t-B)^2}{8B^2}\right)$$

from eq ①.

$$\Pr\left(\left|\sum_{i=1}^{n} X_i\right| \ge t\right) \le 2\exp\left(-\frac{(t-B)^2}{8B^2}\right)$$

Problem 2: Linear regression, 25pts:

Let $(X_1, Y_1), \ldots, (X_n, Y_n) \in \mathbb{R}^d \times \mathbb{R}$ be i.i.d. training data. The Ridge regression solves the following problem:

$$\frac{1}{2}\sum_{j=1}^{n}\left(Y_j - \beta^T X_j\right)^2 + \lambda \sum_{j=1}^{d} \beta_j^2 \to \text{minimize over } \beta = (\beta_1, \ldots, \beta_d)^T \in \mathbb{R}^d,$$

where $\lambda > 0$ is called the "regularization parameter" and the term $\lambda \sum_{j=1}^{d} \beta_j^2$ is called a "penalty."

(1) **(10 pts)** Show that the solution of the Ridge regression is given by

$$\hat{\beta}_\lambda = \left(\mathbf{X}^T\mathbf{X} + \lambda I\right)^{-1}\mathbf{X}^T\mathbf{Y}.$$

Here, I is the $d \times d$ identity matrix, $\mathbf{Y} = (Y_1, \ldots, Y_n)^T$ and $\mathbf{X}$ is a matrix with rows $X_1, \ldots, X_n$. Does the inverse matrix $(\mathbf{X}^T\mathbf{X} + \lambda I)^{-1}$ always exist? One possible approach is to define

$$F(\beta) = \frac{1}{2}\sum_{j=1}^{n}\left(Y_j - \beta^T X_j\right)^2 + \lambda\|\beta\|_2^2$$

and to compute the directional derivative of $F(\beta)$ in direction u:

$$DF(\beta; u) := \lim_{t\to 0}\frac{F(\beta + tu) - F(\beta)}{t}.$$

Then use the fact that $DF(\beta; u) = \langle \nabla F(\beta), u\rangle$ to get the expression for the gradient $\nabla F(\beta)$.

(2) **(15 pts)** If the matrix $\mathbf{X}^T\mathbf{X}$ is not invertible, then $\mathbf{X}^\dagger = \lim_{\lambda\to 0}(\mathbf{X}^T\mathbf{X} + \lambda I)^{-1}\mathbf{X}^T$ is called the Moore-Penrose pseudo-inverse of $\mathbf{X}$. If $\mathbf{X^T X}$ is invertible, then the pseudo-inverse coincides with the usual inverse matrix. Prove that the solution $\hat{\beta}_0$ of the problem

$$\sum_{j=1}^{n}\left(Y_j - \beta^T X_j\right)^2 \to \text{minimize over } \beta = (\beta_1, \ldots, \beta_d)^T \in \mathbb{R}^d$$

given by $\hat{\beta} = \mathbf{X}^\dagger\mathbf{Y}$ has the smallest $\|\cdot\|_2$ - norm among all solutions.

(1) Let $Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} \in \mathbb{R}^n$ & $X = \begin{bmatrix} - X_1 - \\ - X_2 - \\ \vdots \\ - X_n - \end{bmatrix} \in \mathbb{R}^{n\times d}$

Then, the Ridge Regression objective is given by :

$$\hat{\beta} = \min_{\beta\in\mathbb{R}^d} \underbrace{\frac{1}{2}\left(|Y - X\beta|^2 + \lambda|\beta|^2\right)}_{F(\beta)}$$

$$DF(\beta; u) = \lim_{t\to 0}\frac{F(\vec{\beta} + t\vec{u}) - F(\vec{\beta})}{t} = \langle \nabla F(\beta), u\rangle$$

$$= \lim_{t \to 0} \frac{\frac{1}{2}|Y - X\beta - tXu|^2 - \frac{1}{2}|Y - X\beta|^2 + \frac{\lambda}{2}\left(|\beta + ut|^2 - |\beta|^2\right)}{t}$$

$$\frac{1}{2}\left\{ \cancel{|Y - X\beta|^2} + t^2|Xu|^2 - 2t\langle Y - X\beta, Xu\rangle - \cancel{|Y - X\beta|^2} \right\} + \frac{\lambda}{2}\left\{ \cancel{|\beta|^2} + t^2|u|^2 + 2t\langle \beta, u\rangle - \cancel{|\beta|^2} \right\}$$

$$= \lim_{t \to 0} \frac{\qquad\qquad}{t}$$

$$= \lim_{t \to 0} \frac{1}{2}\left\{ t|Xu|^2 - 2\langle Y - X\beta, Xu\rangle \right\} + \frac{\lambda}{2}\left\{ t|u|^2 + 2\langle \beta, u\rangle \right\}$$

$$= \frac{1}{2}\left\{ 2\langle X\beta - Y, Xu\rangle \right\} + \lambda\langle \beta, u\rangle$$

$$= \langle X\beta - Y, Xu\rangle + \lambda\langle \beta, u\rangle$$

$$= \langle X^T(X\beta - Y), u\rangle + \langle \lambda\beta, u\rangle$$

$$= \langle \underbrace{X^T(X\beta - Y) + \lambda\beta}, u\rangle$$

$$= \langle \nabla f(\beta), u\rangle$$

$$\therefore \quad \nabla f(\beta) = X^T(X\beta - Y) + \lambda\beta$$

$$= X^T X\beta - X^T Y + \lambda\beta$$

$$\nabla^2 f = \underbrace{X^T X}_{d \times d} + \lambda I$$

$X^T X$ is PSD since $\forall z \in \mathbb{R}^d$, $z^T X^T X z = (Xz)^T Xz$

$$= |Xz|^2 \geq 0$$

λI is positive definite for $\lambda > 0$.

Hence $X^T X + \lambda I$ is positive definite.

The Hessian $\nabla^2 f = X^T X + \lambda I$ is positive definite.
Hence f is convex. This means that a critical point is a global minima.

Setting gradient to zero, $0 = \nabla f = X^T(X\beta - Y) + \lambda\beta$

$$\Rightarrow \quad 0 = X^T X\beta - X^T Y + \lambda\beta$$

$$\Rightarrow \quad 0 = (X^T X + \lambda I)\beta - X^T Y$$

$$\Rightarrow \quad (X^T X + \lambda I)\beta = X^T Y$$

$$\Rightarrow \quad \hat{\beta}_\lambda = (X^T X + \lambda I)^{-1} X^T Y$$

Since $(X^T X + \lambda I)$ is positive definite, it is invertible.

(2) $$X^\dagger := \lim_{\lambda \to 0} (X^T X + \lambda I)^{-1} X^T$$

From part (1), $\hat{\beta}_\lambda = (X^T X + \lambda I)^{-1} X^T Y$

When $\lambda = 0$ & $X^T X$ is not invertible, there are multiple solutions.

We recall that $\nabla f = X^T(X\beta - Y) + \lambda B$ was set to zero.

$$\Rightarrow \quad X^TX\beta - X^TY + \lambda B = 0$$

$$\Rightarrow \quad (X^TX + \lambda I)\beta = X^TY. \quad \leftarrow \text{Any } \beta \text{ satisfying this is a solution.}$$

Now, $\beta_\lambda = (X^TX + \lambda I)^{-1} X^TY$ when $\lambda > 0$

$$\lim_{\lambda \to 0} \beta_\lambda = \lim_{\lambda \to 0} (X^TX + \lambda I)^{-1} X^TY = X^+Y$$

$$\therefore \quad \beta_0 = X^+Y$$

We want to show that out of all solutions, X^+Y is the minimum norm solution i.e for all β satisfying $(X^TX + \lambda I)\beta = X^TY$

$$X^+ := \lim_{\lambda \to 0} (X^TX + \lambda I)^{-1} X^T$$

Let $X = U\Sigma V^T$ be the SVD of X.

$$X^TX = V\Sigma U^T U\Sigma V^T = V\Sigma^2 V^T$$

$$X^+ = \lim_{\lambda \to 0} (V\Sigma^2 V^T + \lambda I)^{-1} V\Sigma U^T$$

$$= \lim_{\lambda \to 0} (V\Sigma^2 V^T + \lambda V I V^T)^{-1} V\Sigma U^T$$

$$= \lim_{\lambda \to 0} (V(\Sigma^2 + \lambda I)V^T)^{-1} V\Sigma U^T$$

$$= \lim_{\lambda \to 0} (V(\Sigma^2 + \lambda I)^{-1} \underbrace{V^TV}_{I} \Sigma U^T$$

$$= \lim_{\lambda \to 0} V(\Sigma^2 + \lambda I)^{-1}\Sigma U^T$$

$$= \lim_{\lambda \to 0} V(\text{diag}([\cdots \sigma_i^2 + \lambda \cdots]))^{-1}\Sigma U^T$$

$$= \lim_{\lambda \to 0} V\,\text{diag}\left([\cdots \frac{1}{\sigma_i^2 + \lambda} \cdots]\right)\text{diag}([\cdots \sigma_i \cdots])U^T$$

$U\Sigma V^T$
$V\Sigma^{-1}U^T$

$$= \lim_{\lambda \to 0} V\,\text{diag}\left([\cdots \frac{\sigma_i}{\sigma_i^2 + \lambda} \cdots]\right)U^T$$

$$\Rightarrow X^{\dagger} = \lim_{\lambda \to 0} V \begin{bmatrix} \ddots & & 0 \\ & \frac{\sigma_i}{\sigma_i^2 + \lambda} & \\ 0 & & \ddots \end{bmatrix} U^T$$

$$\Rightarrow X^{\dagger} = V \lim_{\lambda \to 0} \begin{bmatrix} \ddots & & \\ & \frac{\sigma_i}{\sigma_i^2 + \lambda} & \\ & & \ddots \end{bmatrix} U^T$$

$f : X \to V^T X U$ f

$g : \lambda \to \left[\frac{\sigma_i}{\sigma_i^2 + \lambda} \right]$

$\because$ f is continuous maps,

$\lim_{\lambda \to 0} f(g(\lambda)) = f(\lim_{\lambda \to 0} g(\lambda))$

If $X^T X = V \Sigma^2 V^T$ is invertible, then Σ^2 has all diagonal entries strictly positive. Hence $\sigma_i^2 > 0 \Rightarrow \sigma_i \neq 0$. Hence $X = U \Sigma V^T$ is invertible too.

Hence $\lim_{\lambda \to 0} \begin{bmatrix} \ddots & \\ & \frac{\sigma_i}{\sigma_i^2 + \lambda} \end{bmatrix} = \begin{bmatrix} \ddots & \\ & \frac{1}{\sigma_i} \end{bmatrix} = \Sigma^{-1}$

$\therefore X^{\dagger} = V \Sigma^{-1} U^T = (U \Sigma V^T)^{-1} = X^{-1}$ when $X^T X$ (and hence X) are invertible.

If $X^T X$ is non-invertible, then $X^T X = V \Sigma^2 V^T$ implies that some diagonal entries of Σ^2 are zero.

Let Z be the index set for which the diagonal entries of Σ^2 are zero, i.e. $\forall i \in Z, \sigma_i^2 = 0$.

$\sigma_i^2 = 0 \Rightarrow \sigma_i = 0$. Hence $X = U \Sigma V^T$ is singular too.

$$X^{\dagger} = V \lim_{\lambda \to 0} \operatorname{diag}\left(\left[\underbrace{\cdots \frac{\sigma_i}{\sigma_i^2 + \lambda}}_{N}, \underbrace{\cdots \frac{\sigma_j \to 0}{\sigma_j^2 + \lambda \to 0}}_{j \in Z} \right] \right) U^T$$

$$\Rightarrow X^{\dagger} = V \lim_{\lambda \to 0} \operatorname{diag}\left(\left[\underbrace{\cdots \frac{\sigma_i}{\sigma_i^2 + \lambda}}_{N}, \underbrace{0, 0, 0 \cdots}_{Z} \right] \right) U^T$$

$$= V \operatorname{diag}\left(\left[\underbrace{\cdots \frac{1}{\sigma_i}}_{N}, 0, 0, \ldots, 0 \right] \right) U^T$$

$$= \sum_{i \in N} \frac{1}{\sigma_i} v_i u_i^T$$ where v_i & u_i are the ith columns of V & U.

$$\beta_0 := \lim_{\lambda \to 0} \beta_\lambda = \lim_{\lambda \to 0} (X^T X + \lambda I)^{-1} X^T Y$$

$$= \sum_{i \in N} \frac{1}{\sigma_i} v_i u_i^T Y$$

We recall that all solutions β satisfy $(X^T X + \lambda I)\beta = X^T Y$.

When $\lambda \to 0$, we have $X^T X \beta = X^T Y$.

Since $X^T X$ is singular, the solution is a particular soln $\vec{p}$ plus the null space of $X^T X = V \Sigma^2 V^T$.

$\{ v_i \mid i \in Z \}$ form a basis of the null space of $X^T X$.

$\therefore$ Any soln $\beta = \vec{p} + \sum_{i \in Z} \alpha_i v_i$ such that $X^T X p = X^T Y$.

Now, β_0 is a particular solution.

$$\therefore \beta = \beta_0 + \sum_{i \in Z} \alpha_i v_i$$

$$\Rightarrow |\beta|^2 = |\beta_0|^2 + \left| \sum_{i \in Z} \alpha_i v_i \right|^2 + 2 \left\langle \beta_0, \sum_{i \in Z} \alpha_i v_i \right\rangle$$

$$= |\beta_0|^2 + \left(\sum_{i \in Z} \alpha_i v_i \right)^T \left(\sum_{j \in Z} \alpha_j v_j \right) + 2 \sum_{i \in Z} \alpha_i \langle \beta_0, v_i \rangle$$

$$= |\beta_0|^2 + \sum_{i,j \in Z} \alpha_i \alpha_j \underbrace{v_i^T v_j}_{\delta_{ij}} + 2 \sum_{i \in Z} \alpha_i \left\langle \sum_{j \in N} \frac{1}{\sigma_j} v_j u_j^T Y, v_i \right\rangle$$

$$= |\beta_0|^2 + \sum_{i \in Z} \alpha_i^2 + 2 \sum_{i \in Z} \alpha_i \sum_{j \in N} \frac{1}{\sigma_j} \underbrace{\langle v_j u_j^T Y, v_i \rangle}_{Y^T u_j v_j^T v_i}$$

$$= |\beta_0|^2 + \sum_{i \in Z} \alpha_i^2 + 2 \sum_{i \in N} \sum_{j \in Z} \alpha_i \cdot \frac{1}{\sigma_j} Y^T u_j \underbrace{v_j^T v_i}_{0, \text{ since } j \in Z \not\ni i \in N}$$

$$= |\beta_0|^2 + \sum_{i \in Z} \alpha_i^2$$

$\therefore$ Any general solution β has norm square given by

$$|\beta|^2 = |\beta_0|^2 + \sum_{i \in Z} \alpha_i^2$$

Since $\alpha_i^2 \geq 0$, the minimum is achieved when $\alpha_i = 0 \ \forall \ i \in Z$

i.e $\underset{\substack{\alpha_i \\ i \in Z}}{\text{argmin}} \ |\beta^2| \Rightarrow \alpha_i = 0 \ \forall \ i \in Z$

$\therefore$ $\min |\beta| = |\beta_0|^2$ & hence β_0 is a min-norm solution & the only one since the rest of the solutions require some $\alpha_i \neq 0$ for $i \in Z$.

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Problem 3, Minkowski functional, 15 points:
Let $K \subset \mathbb{R}^p$ be closed, bounded, convex, symmetric (meaning that $K = -K$), and have non-empty interior (so that K contains some Euclidean ball). Show that the Minkowski functional (gauge)

$$\|x\|_K = \inf\left\{t > 0 : \frac{x}{t} \in K\right\}$$

is a norm.

A norm 'p' has the following properties :-

1) Non-negativity : $p(x) \geq 0$

2) Positive definiteness : $p(x) = 0 \iff x = 0$

3) Homogeneity : $p(sx) = |s| p(x)$ $\forall$ scalars s.

4) Subadditivity/Triangle inequality : $p(x+y) \leq p(x) + p(y)$

1) Check Non-negativity :-

$\|x\|_K = \inf\left\{t > 0 : \frac{x}{t} \in K\right\}$

The infimum is on the set $\{t > 0\}$.
Hence positive by definition.

2) Check Positive definiteness :-

If $x = \vec{0}$, $\frac{x}{t} = \vec{0} \in K$ $\forall$ $t > 0$

Hence $\inf\left\{t > 0 : \frac{x}{t} \in K\right\} = 0$

($\because$ K is non-empty & contains some Euclidean ball which in turn contains the origin $\vec{0}$)

If $\|x\|_K = 0 = t^* = \inf\left\{t > 0 : \frac{x}{t} \in K\right\}$

$\therefore$ $K \ni \lim_{t^* \to 0} \frac{x}{t^*}$ & this is only possible if $x = \vec{0}$.

3) Check Homogeneity :-

First, let us assume $s > 0$.

$$\|sx\|_K = \inf\left\{ t > 0 : \frac{sx}{t} \in K \right\}$$

$$= \inf\left\{ t > 0 : \frac{x}{t/s} \in K \right\}$$

Let $\tau = t/s \Rightarrow \tau s = t$.

$t > 0 \Rightarrow \tau s > 0 \Rightarrow \tau > 0$ since $s > 0$.

$$\|sx\|_K = \inf\left\{ s\tau > 0 : \frac{x}{\tau} \in K \right\}$$

$$= s \inf\left\{ \tau > 0 : \frac{x}{\tau} \in K \right\}$$

$$= s \|x\|_K = |s| \|x\|_K$$

Now assume $s < 0$.

First, we will show that $\|-x\|_K = \|x\|_K$.

$$\|-x\|_K = \inf\left\{ t > 0 : -\frac{x}{t} \in K \right\}$$

Since K is symmetric, $-\frac{x}{t} \in K \Leftrightarrow \frac{x}{t} \in K$.

$$\therefore \|-x\|_K = \inf\left\{ t > 0 : \frac{x}{t} \in K \right\} = \|x\|_K.$$

$$\therefore \|sx\|_K = \underbrace{\|-|s|x\|_K}_{\|-y\|_K} = \underbrace{\||s|x\|_K}_{\|y\|_K} = |s| \|x\|_K$$

from above since $q = |s| > 0$ & $\|qx\|_K = q\|x\|_K$.

If $s = 0$, $\|sx\|_K = \|0\vec{x}\|_K = \|\vec{0}\|_K = 0 = |0| \|x\|_K = |s| \|x\|_K$.

from positive definiteness of $\|\cdot\|_K$

Check Subadditivity / Triangle inequality :-

We will use min instead of inf for a better intuition.

$$\|x\|_K = \min\left\{ t > 0 : \frac{x}{t} \in K \right\}$$

We want to show that $\|x+y\|_K \le \|x\|_K + \|y\|_K$

Let $\|x\|_K = \alpha$. Then $\alpha = \min\{t>0 : \frac{x}{t} \in K\}$

$\Rightarrow \frac{x}{\alpha} \in K$

Similarly if $\|y\|_K = \beta$, then $\frac{y}{\beta} \in K$

Now, $\frac{x+y}{\alpha+\beta} = \frac{\alpha}{\alpha+\beta}\frac{x}{\alpha} + \frac{\beta}{\alpha+\beta}\frac{y}{\beta} \in K$ since K is convex, $\frac{x}{\alpha}, \frac{y}{\beta} \in K$, $\frac{\alpha}{\alpha+\beta} + \frac{\beta}{\alpha+\beta} = 1$, & $\frac{\alpha}{\alpha+\beta}, \frac{\beta}{\alpha+\beta} > 0$ since $\alpha, \beta > 0$

$\Rightarrow \frac{x+y}{\alpha+\beta} \in K$

$\Rightarrow \min\{t>0 : \frac{x+y}{t} \in K\} \le \alpha+\beta$

$\Rightarrow \|x+y\|_K \le \alpha+\beta = \|x\|_K + \|y\|_K$

This argument can be made rigorous for the original definition of Minkowski-gauge as $\|z\|_K = \inf\{t>0 : \frac{z}{t}\}$

Let $\|x\|_K = \alpha = \inf\{t>0 : \frac{x}{t} \in K\}$

By definition of inf as greatest lower bound,

$\forall \epsilon > 0$, $\alpha+\epsilon \in \{t>0 : \frac{x}{t} \in K\} = \{t : t>0\} \cap \{t : \frac{x}{t} \in K\}$

$\Rightarrow \alpha+\epsilon \in \{t : \frac{x}{t} \in K\}$

$\Rightarrow \frac{x}{\alpha+\epsilon} \in K$

Similarly, if $\|y\|_K = \beta$, $\frac{y}{\beta+\epsilon} \in K$.

$\therefore \frac{x+y}{\alpha+\beta+2\epsilon} = \frac{\alpha+\epsilon}{\alpha+\beta+2\epsilon}\frac{x}{\alpha+\epsilon} + \frac{\beta+\epsilon}{\alpha+\beta+2\epsilon}\frac{y}{\beta+\epsilon} \in K$

$\Rightarrow \frac{x+y}{\alpha+\beta+2\epsilon} \in K$

$\Rightarrow \alpha+\beta \in \inf\{\epsilon>0 : \frac{x+y}{\alpha+\beta+2\epsilon} \in K\}$

Now, $\underbrace{\{\alpha+\beta+2\epsilon : \epsilon>0, \frac{x+y}{\alpha+\beta+2\epsilon} \in K\}}_{Q} \subseteq \underbrace{\{t>0 : \frac{x+y}{t} \in K\}}_{R}$

Since $\alpha+\beta = \inf Q \geq \inf R$,

inf over a potentially smaller subset is atmost the inf over the full set.

$$\Rightarrow \alpha+\beta \geq \inf\left\{t>0 : \frac{x+y}{t} \in K\right\}$$

$$\Rightarrow \|x\|_K + \|y\|_K \geq \|x+y\|_K \quad \blacksquare$$

Problem 4, properties of the Gaussian mean width, 20 points):

Required background will be covered on Monday, November 18. Recall the definition of the Gaussian mean width of a bounded set $K \subset \mathbb{R}^p$:

$$w(K) = \mathbb{E} \sup_{z \in K-K} \langle z, g \rangle ,$$

where g has $N(0, I_p)$ distribution and $K-K = \{u-v,\ u,v \in K\}$. Show that

(a) $w(K) = 2\,\mathbb{E}\sup_{z\in K}\langle z, g\rangle$.

(b) $w(K)$ is invariant under affine transformations, meaning that for any $y \in \mathbb{R}^p$ and any $Q \in \mathbb{R}^{p\times p}$ such that $Q^{-1} = Q^T$, $w(QK+y) = w(K)$.

(c) $w(K)$ is invariant with respect to taking the convex hull: if $co(K)$ is the convex hull of K, then $w(\mathrm{co}(K)) = w(K)$.

(d) Let $\mathrm{diam}(K)$ be the diameter of K. Show that

$$\sqrt{\frac{2}{\pi}}\,\mathrm{diam}(K) \leq w(K) \leq \sqrt{p}\,\mathrm{diam}(K).$$

a) $$w(K) = \mathop{\mathbb{E}}_{g\sim N(0,I_p)} \sup_{z\in K-K} \langle z, g\rangle$$

$$= \mathop{\mathbb{E}}_{g} \sup_{u,v\in K} \langle u-v, g\rangle$$

$$= \mathop{\mathbb{E}}_{g} \sup_{u\in K, v\in K} \langle u, g\rangle + \langle v, -g\rangle$$

We have separability of variables over the sup here.

i.e. $\sup_{x,y} f(x) + h(y) = \sup_x f(x) + \sup_y h(y)$

$$\therefore w(K) = \mathop{\mathbb{E}}_{g} \sup_{u\in K} \underbrace{\langle u, g\rangle}_{f(u)} + \sup_{v} \underbrace{\langle v, -g\rangle}_{h(v)}$$

$$\Rightarrow w(K) = \mathbb{E}_g \sup_{u \in K} \langle u, g \rangle + \mathbb{E}_g \sup_{v \in K} \langle v, -g \rangle$$

Since $g \sim N(0, \mathbb{I}_p)$ is symmetric, $-g$ also has the same distribution $N(0, \mathbb{I}_p)$.

$$\therefore \mathbb{E}_g \sup_{v \in K} \langle v, -g \rangle = \mathbb{E}_g \sup_{v \in K} \langle v, g \rangle$$

$$\Rightarrow w(K) = \mathbb{E}_g \sup_{u \in K} \langle u, g \rangle + \mathbb{E}_g \sup_{v \in K} \langle v, g \rangle$$

$$= \mathbb{E}_g \sup_{z \in K} \langle z, g \rangle + \mathbb{E}_g \sup_{z \in K} \langle z, g \rangle$$

$$= 2\, \mathbb{E}_g \sup_{z \in K} \langle z, g \rangle$$

b) Invariance under rigid motion.

Recall, $K \subset \mathbb{R}^p$.

$Q \in \mathbb{R}^{p \times p}$, orthogonal, i.e. $Q^{-1} = Q^T$

$y \in \mathbb{R}^p$

The translated set is $C = QK + y$

$$w(C) = 2\, \mathbb{E}_g \sup_{z \in C} \langle z, g \rangle = 2\, \mathbb{E}_g \sup_{u \in K} \langle Qu + y, g \rangle$$

$$= 2\, \mathbb{E}_g \sup_{u \in K} \left(\langle Qu, g \rangle + \underbrace{\langle y, g \rangle}_{\text{const. w.r.t } z} \right)$$

$$= 2\, \mathbb{E}_g \left(\langle y, g \rangle + \sup_{u \in K} \langle Qu, g \rangle \right)$$

$$= 2\left(\mathop{\mathbb{E}}_{g} \langle y, g \rangle + \mathop{\mathbb{E}}_{g} \sup_{u \in K} \langle Qu, g \rangle \right)$$

$$= 2\left(\langle y, \underbrace{\mathop{\mathbb{E}}_{g} g}_{\vec{0}} \rangle + \mathop{\mathbb{E}}_{g} \sup_{u \in K} \langle Qu, g \rangle \right)$$

$$= 2 \mathop{\mathbb{E}}_{g} \sup_{u \in K} \langle Qu, g \rangle = 2 \mathop{\mathbb{E}}_{g} \sup_{u \in K} \langle u, Q^T g \rangle$$

Since $g \sim N(0, I_p)$ is rotationally invariant, $Q^T g$ has the same distribution $N(0, I_p)$.

$$\mathop{\mathbb{E}}_{g}[Q^T g] = Q^T \mathbb{E}[g] = Q^T \vec{0} = \vec{0} = \mathop{\mathbb{E}}_{g}[g]$$

$$\mathrm{Var}(Q^T g) = \mathop{\mathbb{E}}_{g}[Q^T g (Q^T g)^T] = \mathop{\mathbb{E}}_{g} Q^T g g^T Q = Q^T \mathbb{E}[g g^T] Q$$
$$= Q^T \mathrm{Var}(g) Q$$
$$= Q^T I_p Q$$
$$= Q^T Q = I_p = \mathrm{Var}(g)$$

Also gaussian remains gaussian under linear transform. So $Q^T g$ is gaussian and has same mean & variance as g. Hence $Q^T g$ has same distribution as g, which is $N(0, I_p)$.

$$\therefore w(C) = 2 \mathop{\mathbb{E}}_{g} \sup_{u \in K} \langle u, Q^T g \rangle = 2 \mathop{\mathbb{E}}_{g} \sup_{u \in K} \langle u, g \rangle = w(K)$$

The above argument can also be made rigorous measure theoretically.

$$\therefore \quad w(C) = w(QK + y) = w(K).$$

c) Invariance under Convexification by Convex Hull.

The convex hull of K is $Co(K) = \{ \alpha x + (1-\alpha) y \mid x, y \in K, \alpha \in [0,1] \}$

We will show $w(K) \leq w(Co(K))$

$$w(K) = 2 \mathbb{E}_g \sup_{z \in K} \langle z, g \rangle$$

$$w(Co(K)) = 2 \mathbb{E}_g \sup_{z \in Co(K)} \langle z, g \rangle$$

We will show that $\forall g \quad \sup_{z \in K} \langle z, g \rangle = \sup_{z \in Co(K)} \langle z, g \rangle$

Since $K \subseteq Co(K)$, $\sup_{z \in Co(K)} \langle z, g \rangle \geq \sup_{z \in K} \langle z, g \rangle \rightarrow (1)$

Now we will show the reverse $\sup_{z \in K} \langle z, g \rangle \leq \sup_{z \in Co(K)} \langle z, g \rangle$

By Caratheodory theorem, if $z \in Co(K)$, then $\exists\, z_1, z_2 \ldots z_{p+1}$ such that $z = \sum_{i=1}^{p+1} \alpha_i z_i$ with $\alpha_i \in [0,1]$ & $\sum_{i=1}^{p+1} \alpha_i = 1$

Let $z_0 \in Co(K)$ be a given point.

By Caratheodory Theorem, $z_0 = \sum_{i=1}^{p+1} \alpha_i z_i$ where $z_i \in K$

For any $g \in \mathbb{R}^p$, $\langle z_0, g \rangle = \langle \sum_{i=1}^{p+1} \alpha_i z_i, g \rangle$

$$\Rightarrow \langle z_0, g \rangle = \sum_{i=1}^{p+1} \alpha_i \langle z_i, g \rangle$$

Let $M = \sup_{z \in K} \langle z, g \rangle$. Clearly $\forall i$, $\langle z_i, g \rangle \leq M$.

$$\therefore \langle z_0, g \rangle = \sum_{i=1}^{p+1} \alpha_i \langle z_i, g \rangle \leq \sum_{i=1}^{p+1} \alpha_i M = M \sum_{i=1}^{p+1} \alpha_i = M.$$

$$\therefore \quad \langle z_0, g \rangle \leq M \quad \forall \; z_0 \in Co(K)$$

$$\Rightarrow \quad \sup_{z_0 \in Co(K)} \langle z_0, g \rangle \leq M = \sup_{z \in K} \langle z, g \rangle \longrightarrow (2)$$

From (1) & (2) :-

$$\sup_{z \in Co(K)} \langle z, g \rangle = \sup_{z \in K} \langle z, g \rangle$$

$$\Rightarrow 2 \, \mathbb{E}_g \sup_{z \in Co(K)} \langle z, g \rangle = 2 \, \mathbb{E}_g \sup_{z \in K} \langle z, g \rangle$$

$$\Rightarrow \quad w(Co(K)) = w(K)$$

d) To show :-

$$\sqrt{\frac{2}{\pi}} \, diam(K) \leq w(K) \leq \sqrt{p} \, diam(K)$$

$$diam(K) := \sup \{ |x - y| \mid x, y \in K \}$$

$$= \sup_{x, y \in K} |x - y| = \sup_{z \in K - K} |z|$$

$$w(K) = \mathbb{E}_g \sup_{z \in K - K} \langle z, g \rangle \leq \mathbb{E}_g \sup_{z \in K - K} |z| |g|$$

↓ Cauchy-Schwarz inequality

$$= \mathbb{E}_g |g| \sup_{z \in K - K} |z|$$

$$= \mathop{\mathbb{E}}_{g} |g| \, \text{diam}(K) = \text{diam}(K) \mathop{\mathbb{E}}_{g} |g|$$

Now, $\mathop{\mathbb{E}}_{g} |g| = \mathop{\mathbb{E}}_{g} \sqrt{|g|^2} = \mathop{\mathbb{E}}_{g} \sqrt{g^T g} \leq \sqrt{\mathop{\mathbb{E}}_{g} g^T g}$

Jensen's Inequality
$\sqrt{\cdot}$ is concave

$$\Rightarrow \mathop{\mathbb{E}}_{g} |g| \leq \sqrt{\mathop{\mathbb{E}}_{g} \text{tr}(g^T g)} = \sqrt{\mathop{\mathbb{E}}_{g} \text{tr}(g g^T)}$$

$$= \sqrt{\text{tr}\left(\mathop{\mathbb{E}}_{g} g g^T\right)} = \sqrt{\text{tr}\left(\text{Var}(g)\right)}$$

$$= \sqrt{\text{tr}(I_p)} = \sqrt{p}$$

Alternatively $\mathop{\mathbb{E}}_{g} |g| = \mathop{\mathbb{E}}_{g} \sqrt{|g|^2} \leq \sqrt{\mathop{\mathbb{E}} |g|^2}$

$$\Rightarrow \mathop{\mathbb{E}}_{g} |g| = \sqrt{\mathop{\mathbb{E}}_{g} \sum_{i=1}^{p} g_i^2} = \sqrt{\sum_{i=1}^{p} \mathop{\mathbb{E}}_{g_i} g_i^2}$$

$$\Rightarrow \mathop{\mathbb{E}}_{g} |g| = \sqrt{\sum_{i=1}^{p} \text{Var}(g_i)} = \sqrt{\sum_{i=1}^{p} 1}$$

$$\therefore \mathop{\mathbb{E}}_{g} |g| \leq \sqrt{p}$$

$$\therefore w(K) = \text{diam}(K) \mathop{\mathbb{E}}_{g} |g| \leq \sqrt{p} \, \text{diam}(K)$$

Now, lets show the other inequality

$$\sqrt{\frac{2}{\pi}}\,\mathrm{diam}(K) \le w(K)$$

$$w(K) = \mathbb{E}_g \sup_{z \in K-K} \langle z, g \rangle$$

Let $u \in K-K$. Since $K-K$ is symmetric, $-u \in K-K$.

For any arbitrary $u \in K-K$, & fixed g, by definition of Supremum, we have $\sup_{z \in K-K} \langle z, g \rangle \ge \langle u, g \rangle$

By the same argument, since $-u \in K-K$,

$$\sup_{z \in K-K} \langle z, g \rangle \ge \langle -u, g \rangle = -\langle u, g \rangle$$

$$\therefore \sup_{z \in K-K} \langle z, g \rangle \ge \max\{\langle u, g \rangle, -\langle u, g \rangle\}$$

$$\Rightarrow \sup_{z \in K-K} \langle z, g \rangle \ge |\langle u, g \rangle| \quad \forall u \in K-K \text{ \& } \forall g \sim \mathcal{N}(0, \mathbb{I}_p)$$

$M(g)$ ↓ random variable as g is random

$Y(g)$

$$\Rightarrow \underbrace{\mathop{\mathbb{E}}_{g} M(g)}_{w(K)} \geq \mathop{\mathbb{E}}_{g} Y(g)$$

$$\Rightarrow w(K) \geq \mathop{\mathbb{E}}_{g} |\langle u, g\rangle|$$

Since g is gaussian, $\langle u, g\rangle$ is also gaussian since it is a linear transform $u^T g$ of g.

$$\mathbb{E}[\langle u, g\rangle] = \langle u, \mathbb{E}[g]\rangle = \langle u, 0\rangle = 0$$

$$\begin{aligned}
\mathrm{Var}(\langle u, g\rangle) &= \mathop{\mathbb{E}}_{g}((\langle u, g\rangle - 0)(\langle u, g\rangle - 0)) \\
&= \mathop{\mathbb{E}}_{g}(u^T g g^T u) \\
&= u^T \mathop{\mathbb{E}}_{g}[g g^T] u \\
&= u^T \Pi_p u = u^T u = \|u\|^2
\end{aligned}$$

Hence, $\langle u, g\rangle$ has the same distribution as $N(0, \|u\|^2)$

or $\langle u, g\rangle \overset{\text{dist.}}{\sim} |u| h$ where $h \sim N(0,1)$

$$\therefore \mathop{\mathbb{E}}_{g} |\langle u, g\rangle| = \mathop{\mathbb{E}}_{h} |u| |h| = |u| \mathop{\mathbb{E}}_{h} |h|$$

$$\mathop{\mathbb{E}}_{h} |h| = \int |h| p(h)\, dh = \int_{-\infty}^{\infty} |h| \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}h^2} dh$$

even function of h.

$$= \frac{1}{\sqrt{2\pi}} \cdot 2 \cdot \int_0^{\infty} |h| e^{-\frac{1}{2}h^2} dh$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty h\, e^{-\frac{1}{2}h^2}\, dh$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-t}\, dt$$

$t = \frac{1}{2}h^2$, $dt = h\,dh$

$h \to 0$, $t \to 0$

$h \to \infty$, $t \to \infty$

$$= \sqrt{\frac{2}{\pi}} \left[-e^{-t} \right]_0^\infty$$

$$= \sqrt{\frac{2}{\pi}} \left[-e^{-\infty} - (-e^{-0}) \right] = \sqrt{\frac{2}{\pi}} \left[0 + 1 \right]$$

$$= \sqrt{\frac{2}{\pi}}$$

$$\therefore \mathop{\mathbb{E}}_{g} \langle u, g \rangle = |u| \mathop{\mathbb{E}}_{h} |h| = |u| \sqrt{\frac{2}{\pi}}$$

Hence,

$$w(K) \geq \mathop{\mathbb{E}}_{g} |\langle u, g \rangle| = |u| \sqrt{\frac{2}{\pi}} \quad \forall\, u \in K - K$$

$$\Rightarrow w(K) \geq \sup_{u \in K-K} |u| \sqrt{\frac{2}{\pi}}$$

$$\Rightarrow w(K) \geq \sqrt{\frac{2}{\pi}} \underbrace{\sup_{u \in K-K} |u|}_{\text{diam}(K)}$$

$$\Rightarrow w(K) \geq \sqrt{\frac{2}{\pi}}\, \text{diam}(K)$$

$$\therefore \sqrt{\frac{2}{\pi}}\, \text{diam}(K) \leq \sqrt{\frac{2}{\pi}}\, \text{diam}(K) \leq \sqrt{p}\, \text{diam}(K)$$

Problem 5, bonus: matrix inverse, 10 points: Let A be a positive definite matrix. Prove that the map $A \mapsto A^{-1}$ is convex, that is, for any positive definite A, B and any $0 \le \lambda \le 1$,

$$(\lambda A + (1-\lambda)B)^{-1} \preceq \lambda A^{-1} + (1-\lambda)B^{-1}.$$

(Hint – use the following fact without the proof: the block matrix $\begin{pmatrix} T & C \\ C^T & M \end{pmatrix}$ is nonnegative definite if and only if $C^T T^{-1} C \preceq M$; the matrix $C^T T^{-1} C - M$ is known as the Schur complement. Now apply it (how?) with $T = A$, $M = A^{-1}$, $C = I$ and $T = B$, $M = B^{-1}$, $C = I$)

We want to show that if A & B are positive definite, i.e. $A, B \succ 0$, then the map $A \to A^{-1}$ is convex.

Let $A, B \succ 0$.

$$\begin{bmatrix} A & I \\ I & A^{-1} \end{bmatrix} \succcurlyeq 0 \quad \text{iff} \quad I^T A^{-1} I \le A^{-1}$$

$$\text{or} \quad \underbrace{A^{-1} \le A^{-1}}_{\text{true}}$$

$$\Downarrow$$

$$\begin{bmatrix} A & I \\ I & A^{-1} \end{bmatrix} \succcurlyeq 0$$

Similarly, $\begin{bmatrix} B & I \\ I & B^{-1} \end{bmatrix} \succcurlyeq 0$

$$\therefore \lambda \begin{bmatrix} A & I \\ I & A^{-1} \end{bmatrix} + (1-\lambda) \begin{bmatrix} B & I \\ I & B^{-1} \end{bmatrix} \succcurlyeq 0 \quad \forall \lambda \in [0,1]$$

$$\Rightarrow \begin{bmatrix} \lambda A + (1-\lambda)B & I \\ I & \lambda A^{-1} + (1-\lambda)B^{-1} \end{bmatrix} \succcurlyeq 0$$

$$\Rightarrow I^T (\lambda A + (1-\lambda)B)^{-1} I \le \lambda A^{-1} + (1-\lambda)B^{-1}$$

$$\Rightarrow (\lambda A + (1-\lambda)B)^{-1} \le_{\mathbb{R}^n_{++}} \lambda A^{-1} + (1-\lambda)B^{-1}$$

↳ convexity w.r.t the standard positive cone $\mathbb{R}^n_{++}$